Solomonoff Axiom Induction:
Definitions, Basic Theorems, and What a Theory Is and Does

Kaarel Hänni Claude (Anthropic)*

October 2026

Abstract

(To be written.)

Contents

1 Introduction **2**
2 Setting **2**
3 Basic theorems **2**
4 Relation to Solomonoff operator induction **2**
5 Logic: random theories, Π_1 truth and Gödel phenomena **2**
6 What a theory does and what it is **2**
7 Theory shape relative to a reasoner **2**
8 Theoretical terms: Craig, Ramsey and the theoretician's dilemma **2**
9 Approximate laws and the best system **2**
10 A worked example: Kepler and Newton **2**
11 Limitations **3**
12 Related work **3**
13 Discussion **3**
A Proofs for the setting and the basic theorems **3**
B Proofs for the relation to operator induction **3**
C Proofs for random theories, Π_1 data and Gödel phenomena **3**
D Proofs for what a theory does and is **3**
E Proofs for theory shape **3**
F Proofs for theoretical terms **3**
G Proofs for approximate laws and the best system **4**
H The Kepler computations **4**

*Claude is an AI system made by Anthropic. Claude produced the proofs, the computations and the text, under Kaarel Hänni's direction and starting from his questions and working notes (Hänni, 2026). The results were checked by separate Claude instances acting as referees, which re-derived the proofs and re-implemented the computations with independent code. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant.

I Notation **4**

1 Introduction

(To be written.)

2 Setting

(To be written.)

3 Basic theorems

(To be written.)

4 Relation to Solomonoff operator induction

(To be written.)

5 Logic: random theories, Π_1 truth and Gödel phenomena

(To be written.)

6 What a theory does and what it is

(To be written.)

7 Theory shape relative to a reasoner

(To be written.)

8 Theoretical terms: Craig, Ramsey and the theoretician's dilemma

(To be written.)

9 Approximate laws and the best system

(To be written.)

10 A worked example: Kepler and Newton

(To be written.)

11 Limitations

(To be written.)

12 Related work

(To be written.)

13 Discussion

(To be written.)

References

Kaarel Hänni. Working notes, 2023–2026. Public repository, `https://github.com/kaarelh/notes`, 2026. Accessed October 2026. The note on Solomonoff axiom induction is in the folder `ai/learning/solomonoff induction`.

A Proofs for the setting and the basic theorems

(To be written.)

B Proofs for the relation to operator induction

(To be written.)

C Proofs for random theories, Π_1 data and Gödel phenomena

(To be written.)

D Proofs for what a theory does and is

(To be written.)

E Proofs for theory shape

(To be written.)

F Proofs for theoretical terms

(To be written.)

G Proofs for approximate laws and the best system

(To be written.)

H The Kepler computations

(To be written.)

I Notation

(To be written.)